

Jiannan LI

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Current position

Assistant Professor

School of Computing and Information Systems, Singapore Management University

Research Interest

Human-AI Interaction, Human-Robot Interaction

Education

2016-2023	PhD in Computer Science, University of Toronto
2012-15	MSc in Computer Science, University of Calgary
2008-12	BENG in Automation and Control, Southeast University, China

Publications

PEER-REVIEWED CONFERENCE AND JOURNAL ARTICLES

- DIS[C.21] Tianyi Zhang, Emran Poh, Yueyue Hou, Yi-Chieh Lee, Renwen Zhang, Jiannan Li, Anthony Tang. 2026. **“Grandpa, Can You Speak Nicer?”: Envisioned Chatbot Roles and Design Tensions in Intergenerational Communication Conflicts.** In *Proceedings of the 2026 Designing Interactive Systems Conference*.
- DIS[C.20] ShunYi Yeo, Tianyi Zhang, Scott Bateman, Gary Hsieh, Young-Ho Kim, Simon Tangi Perrault, Jiannan Li, Anthony Tang. 2026. **Group Conversational Agents: A Review of Designs that Support and Shape Group Interaction.** In *Proceedings of the 2026 Designing Interactive Systems Conference*. 🏆 Honorable Mention Award (top 5%).
- TVCG[J.7] Shaolun Ruan, Feng Liang, Rohan Ramakrishna, Chao Ren, Rudai Yan, Qiang Guan, Jiannan Li, Yong Wang. 2026. **Towards Explainable Quantum AI: Informing the Encoder Selection of Quantum Neural Networks via Visualization.** In *IEEE Transactions on Visualization and Computer Graphics*.
- SenSys[C.19] Changshuo Hu, Hung Manh Pham, Ting Dang, Jiannan Li, Rajesh Balan, Dong Ma. 2026. **From Cheap to Chic: Enhancing Music Playback Quality of Budget Earphones via Hardware-Aware Learning.** In *Proceedings of the 2026 ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems*.
- CHI[C.18] Emran Poh, Yueyue Hou, Tianyi Zhang, Jiannan Li. 2026. **‘Show It, Don’t Just Say It’: The Complementary Effects of Instruction Multimodality for Software Guidance.** In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*.

- ICHMS[C.17] Ruochen Li, Ziyi Chang, Junyan Hu, Jiannan Li, Amir Atapour-Abarghouei, Hubert PH Shum. 2026. **ART: Adaptive Relational Transformer for Pedestrian Trajectory Prediction with Temporal-Aware Relations**. In *The 6th IEEE International Conference on Human-Machine Systems*.
- TVCG[J.6] Manusha Karunathilaka, Litian Lei, Yiming Gao, Yong Wang, Jiannan Li. 2026. **Compendia: Automated Visual Storytelling Generation from Online Article Collection**. In *IEEE Transactions on Visualization and Computer Graphics*.
- TVCG[J.5] Shaolun Ruan, Rui Sheng, Xiaolin Wen, Jiachen Wang, Tianyi Zhang, Yong Wang, Tim Dwyer, Jiannan Li. 2025. **Qualitative Study for LLM-assisted Design Study Process: Strategies, Challenges, and Roles**. In *IEEE Transactions on Visualization and Computer Graphics*. 🏆 Honorable Mention Award (top 5%) at VIS 25.
- IMWUT[J.4] Ruyi Li, Jingfei Guo, Xinyi Zhang, Xuji Zhang, Zeqing Li, Jiannan Li, Jiangtao Gong. 2025. **Guiding Multiple Remote Users in Physical Tasks with Language-driven Robotic Telepresence**. In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*.
- CSCW[J.3] Qian Wan, Jiannan Li, Huanchen Wang, Zhicong Lu. 2025. **Polymind: Parallel Visual Diagramming with Large Language Models to Support Prewriting Through Microtasks**. In *Proceedings of the ACM on Human-Computer Interaction, CSCW '25*.
- CHI[C.16] Tianyi Zhang, Colin Au Yeung, Emily Aurelia, Yuki Onishi, Neil Chulpongsatorn, Jiannan Li, Anthony Tang. 2025. **Prompting an Embodied AI Agent: How Embodiment and Multi-modal Signaling Affects Prompting Behaviour**. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 🏆 Honorable Mention Award (top 5%).
- HRI[C.15] Karthik Mahadevan, Blaine Lewis, Jiannan Li, Bilge Mutlu, Anthony Tang, Tovi Grossman. 2025. **ImageInThat: Manipulating Images to Convey User Instructions to Robots**. In *Proceedings of the 2025 IEEE/ACM International Conference on Human-Robot Interaction*. 🏆 Best Paper Award (top 1%).
- UIST[C.14] Ludwig Sidenmark, Tianyu Zhang, Leen Al Labadidi, Jiannan Li, Tovi Grossman. 2024. **Desk2Desk: Optimization-based Mixed Reality Workspace Integration for Remote Side-by-side Collaboration**. In *Proceedings of the 2024 Annual ACM Symposium on User Interface Software and Technology*. Acceptance rate 24%.
- DIS[C.13] Setareh Aghel Manesh, Tianyi Zhang, Yuki Onishi, Kotaro Hara, Scott Bateman, Jiannan Li, and Anthony Tang. 2024. **How People Prompt to Create Interactive VR Scenes**. In *Proceedings of the 2024 ACM Conference on Designing Interactive Systems*. Acceptance rate 27.4%.
- CHI[C.12] Jiannan Li, Maurício Sousa, Karthik Mahadevan, Bryan Wang, Paula Akemi Aoyaui, Nicole Yu, Angela Yang, Ravin Balakrishnan, Anthony Tang, and Tovi Grossman. 2023. **Stargazer: An Interactive Camera Robot for Capturing How-To Videos Based on Subtle Instructor Cues**. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. Acceptance rate 28.4%.
- VR[C.11] Sixuan Wu, Jiannan Li, Mauricio Sousa, and Tovi Grossman. 2023. **Investigating Guardian Awareness Techniques to Promote Safety in Virtual Reality**. In *Proceedings of the 30th IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR '23)*.
- CSCW[J.2] Kartikaeya Kumar, Lev Poretski, Jiannan Li, and Anthony Tang. 2022. **Tourgether360: Collaborative Exploration of 360° Videos Using Pseudo-Spatial Navigation**. In *Proceedings of the ACM on Human Computer Interaction*.
- CHI[C.10] Kevin Huang, Jiannan Li, Maurício Sousa, and Tovi Grossman. 2022. **immersivePOV: Filming How-To Videos with a Head-Mounted 360° Action Camera**. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 🏆 Honorable Mention Award (top 5%).

- CHI[C.9] Jiannan Li, Maurício Sousa, Chu Li, Jessie Liu, Yan Chen, Ravin Balakrishnan, and Tovi Grossman. 2022. **ASTEROIDS: Exploring Swarms of Mini-Telepresence Robots for Physical Skill Demonstration**. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* Accepted with minor revision, acceptance rate 12.5%.
- UIST[C.8] Jiannan Li, Jiahe Lyu, Maurício Sousa, Ravin Balakrishnan, Anthony Tang, and Tovi Grossman. 2021. **Route Tapestries: Route Tapestries: Navigating 360 Virtual Tour Videos Using Slit-Scan Visualizations**. In *Proceedings of the 34th Annual ACM Symposium on User Interface Software and Technology*. Acceptance rate 25.9%.
- UIST[C.7] Jiangtao Gong, Teng Han, Siling Guo, Jiannan Li, Siyu Zha. 2021. **HoloBoard: an Immersive Teaching Board for Lecture-based Classes**. In *Proceedings of the 34th Annual ACM Symposium on User Interface Software and Technology*. Acceptance rate 25.9%.
- CHI[C.6] Zhicong Lu, Chenxinran Shen, Jiannan Li, Hong Shen, and Daniel Wigdor. 2021. **More Kawaii than a Real-Person Live Streamer: Understanding How the Otaku Community Engages with and Perceives Virtual YouTubers**. In *Proceedings of 2021 SIGCHI Conference on Human Factors in Computing Systems*. Acceptance rate 26.3%.
- GI[C.5] Jiannan Li, Ravin Balakrishnan, and Tovi Grossman. 2020. **StarHopper: A Touch Interface for Remote Object-Centric Drone Navigation**. In *Proceedings of the 2020 Graphical Interface Conference*.
- CHI[C.4] Teng Han, Jie Liu, Khalad Hasan, Mingming Fan, Junhyeok Kim, Jiannan Li, Xiangmin Fan, Feng Tian, Edward Lank, and Pourang Irani. 2019. **PinchList: Leveraging Pinch Gestures for Hierarchical List Navigation on Smartphones**. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Acceptance rate 23.8%.
- CHI[C.3] Teng Han, Jiannan Li, Khalad Hasan, Keisuke Nakamura, Randy Gomez, Ravin Balakrishnan, and Pourang Irani. 2018. **PageFlip: Leveraging Page-Flipping Gestures for Efficient Command and Value Selection on Smartwatches**. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. Acceptance rate 25.7%.
- IJHCS[J.1] Jiannan Li, Saul Greenberg, Ehud Sharlin. 2017. **A Two-Sided Collaborative Transparent Display Supporting Workspace Awareness**. In *International Journal of Human Computer Studies*, 101:23-44.
- DIS[C.2] Jiannan Li, Saul Greenberg, Ehud Sharlin, and Joaquim Jorge. 2014. **Interactive two-sided transparent displays: designing for collaboration**. In *Proceedings of the 2014 conference on Designing interactive systems*. Acceptance rate 27%.
- ICCSNT[C.1] Jiannan Li, Pengbo Li, Hongyi Liu, Dong Li, and Jian Tang. 2011. **A Contactless Battery Charging and Monitoring System for Wireless Sensor Network Nodes**. In *Proceedings of 2011 International Conference on Computer Science and Network Technology*.

PRE-PRINTS

- arXiv Rahul Arora, Jiannan Li, Gongyi Shi, and Karan Singh. 2021. **Thinking Outside the Lab: VR Size & Depth Perception in the Wild**. arXiv:2105.00584.

POSTERS & WORKSHOPS

- CHILBW Nikhita Joshi, Richard Li, Jiannan Li, Leonardo Pavanatto, Michel Pahud, Jatin Sharma, Bongshin Lee, Hugo Romat, William A.S. Buxton, Nicolai Marquardt, Ken Hinckley, Nathalie Riche.

Opportunistic Nudges for Task Migration Between Personal Devices. In *CHI '24 Extended Abstracts on Human Factors in Computing Systems*

AIVR Zhuoyue Lyu, Jiannan Li, Bryan Wang. 2021. **ALive: Interactive Visualization and Sonification of Neural Networks in Virtual Reality.** In *2021 IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR).*

HAI Jiannan Li, Maurício Sousa, Ravin Balakrishnan, and Tovi Grossman. 2021. **Constellation: a Multi-User Interface for Remote Drone Tours.** In *Proceedings of the 9th International Conference on Human-Agent Interaction (HAI '21).*

CHIEA Jiannan Li, Ehud Sharlin, Saul Greenberg, and Michael Rounding. 2013. **Designing the car iWindow: exploring interaction through vehicle side windows.** In *CHI '13 Extended Abstracts on Human Factors in Computing Systems.*

Industry Employment

2022 Research Intern at Microsoft Research, Redmond, US
 2015-16 Software engineer at Nureva Inc., Calgary, Canada
 2014 Research Intern at Lenovo Research, Beijing, China

Grants

2026-2027 Ministry of Education Academic Research Fund Tier 1: **Towards Memory-Augmented Multi-modal Reasoning for Ambiguity-Resilient Human–Robot Interaction** (Co-PI, 150,000 SGD)
 2025- Google Gift: **Literacy-Adaptive Interaction Primitives for Human-Agent Interaction with Older Adults** (PI, 35,000 USD)
 2025-2026 SMU-Durham Seedcorn Fund for Human-Robot Interaction (PI, 12,500 USD)
 2023-2025 Ministry of Education Academic Research Fund Tier 1: **Developing Intelligent Agents that Assist with Human Situational Difficulties** (PI, 119,333 SGD)

Awards

2026 Best Paper Honourable Mention, DIS 2026
 2025 Best Paper Honourable Mention, VIS 2025
 2025 Best Paper Honourable Mention, CHI 2025
 2025 Best Paper Award, HRI 2025
 2022 Best Paper Honourable Mention, CHI 2022
 2022 Snap Creative Challenge Award
 2021 Wolfond Scholarship in Wireless Information Technology (5,000 CAD)
 2016-20 Ontario Trillium Scholarship (40,000 CAD per year)
 2016-18 Wolfond Scholarship in Wireless Information Technology (20,000 CAD)
 2013,14 University of Calgary Computer Science Graduate Research Award (2,000 CAD)

Services

COMMITTEE MEMBER

2026	Interactivity Chair, ACM DIS
2025,26	Technical Program Committee member, ACM UIST
2024-26	Technical Program Committee member, ACM CHI
2022-24	Technical Program Committee member, ACM ICM
2021	Web Chair, IEEE VR

REVIEWER

2017,20-24	ACM CHI, special recognition for outstanding reviews for CHI '22
2020,22-24	ACM UIST, special recognition for outstanding reviews for UIST '22 (2)
2020,22-24	ACM CSCW, special recognition for outstanding reviews for CSCW '23 and '24
2023,24	ACM IMWUT
2021,22,24	IEEE VR
2024	IEEE ISMAR
2023,24	ACM DIS
2024	ACM CHI PLAY
2022	ACM VRST
2014,15,22	ACM TEI
2016,17	IEEE RO-MAN
2025	ACM Transactions on Human-Robot Interaction
2023	ACM Transactions on Computer-Human Interaction
2021	IEEE Transactions on Human-Machine Systems
2020	International Journal on Human-Computer Studies

Supervision & Mentoring

PHD STUDENT

2023-	Tianyi Zhang (SMU, co-advised with Tony Tang)
2024-	Yueyue Hou (SMU)
2024-	Manusha Karunathilaka (SMU, co-advised with Wang Yong)
2024-2026	Shaolun Ruan (SMU, co-advised with Wang Yong), Google PhD Fellowship
2025-	Emran Poh (SMU)

MASTER STUDENT

2023	Setareh Aghel Manesh (SMU, visiting student from UCalgary), [C.13]
2022	Paula Akemi Aoyagui (UToronto), [C.12]
2022	Nicole Sin Ting Yu (UToronto), [C.12]
2021	Chu Li (UToronto), [C.9]

UNDERGRADUATE STUDENT

2022	Angela Yang (UToronto), [C.12]
2021,22	Sixuan Wu (UToronto), [C.11]
2021,22	Yuchen Fan (UToronto)
2021	Kartikaeya Kumar (IIT Guwahati), [J.2]
2021	Kevin Huang (UToronto), [C.10]
2020,21	Jiahe Lyu (UToronto), [C.8]

Teaching

2024-26	IS133 (Web Application Development)
2025	CS616 (Human-AI Interaction Design)

References

Available upon request.